



### Termites build massive landfill

Termites have built massive dumps on empty grounds in Brazil over thousands of years as deposition sites. <https://digit.in/dec18-66-1>



### Rare millipede found

A recently discovered millipede, relatively "new to science", has only ever been found thrice - all in south Wales. <https://digit.in/dec18-66-2>

# THE DUST CLOUD MOONS

After over six decades of searching, scientists have finally confirmed the existence of dust cloud satellites around the Earth

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ince the 19th century, scientists have been looking for a second moon of the Earth.

Over the years, there have been multiple claims and sightings of additional satellites of the Earth. In the 1960s, John Bargby claimed to have spotted as many as ten previously undiscovered satellites of the Earth. These claims were never confirmed.

Before going any further, we need to understand a tiny slice of celestial mechanics. A pair of bodies, such as the Sun-Earth system, or the Earth-Moon system, have five special points in space around them. The combined gravitational influence of the Earth and the Sun, keep any objects located in points, in the same relative position. Essentially, anything that finds its way to these points, tend to remain there.

These positions are called Lagrangian points, and a system of any two bodies has five of them, labeled L1, L2, L3, L4 and L5. In the Earth-Moon system, the L4 and L5 points are not entirely stable, and are constantly shifting slightly, because of the gravitational pull of the Sun. Objects in these points may be under the gravitational influence of other planets as well. However, any objects at these locations would be ideal to find any additional satellites of the Earth.

In the 1950s, Polish astronomer Kazimierz Kordylewski began a search for "Trojan" satellites of the Earth. The L4 and L5 points are together known as the Trojan points, and a number of other planets have satellites here. Jupiter has more than 6,000 trojan satellites. There are seventeen Neptunian trojans identified, and four Martian trojans. There are a number of trojan

moons in the Saturnian system. In the 1950s, Kordylewski started looking for Earth trojans. In 1956, Kordylewski spotted a faint, diffuse patch of light at one of the trojan points, with his naked eye. This was indicative of a cloud of dust. Almost five years later, Kordylewski managed to capture the dust cloud on film. In 1966, astronomers reported seeing two faint patches at both the L4 and L5 points while flying on board a NASA plane. In 1975, the Orbiting Solar Observatory Announced the discovery of what were then called Kordylewski Dust Clouds (KDC), after a 15 month long observation campaign. The faint dust clouds were subsequently photographed, but the cloud satellites remained elusive. Then there were a number of observation campaigns where astronomers specifically looked at these points and found nothing. In 2009, the Japanese Hiten probe